Venkata Lakshmi Parimala Pasupuleti

AI Engineer · LangChain · LangGraph · MCP · RAG Pipelines · Agentic Workflows · Tool Integration · LLM Evaluation · Python · FastAPI · AWS

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PROFESSIONAL SUMMARY

AI Engineer with 8+ years production SWE experience and 2 years hands-on agentic AI - exactly the profile Chamberlain's JD specifies. Built LeaseI: 8-agent LangGraph system with MCP, tool integration, durable state management, and conditional HITL routing. Built AgentProbe: open-source AI evaluation infrastructure with trace-level observability, 5-metric eval engine, and CI-integrated regression harness. Strong on agent reasoning patterns, RAG pipelines, and LLM orchestration frameworks (LangChain, LangGraph). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and data pipelines; surfacing insights from raw business data into application-facing APIs and reporting layers.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern

- Built agentic system (LangGraph stateful graph + FastAPI): parse → plan → retrieve → validate → conditional routing — RAG over regulatory corpus, validation guardrails enforced in code before output export.

George Mason University Aug 2024 – Aug 2025

Research Associate

- Built ML pipelines (Python, R) across 190+ countries (R^2=0.977); fine-tuned HuggingFace classification model with PyTorch; deployed inference API on Quart; Isolation Forest anomaly detection across 3 research pipelines.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **8+ years Python production SWE** — owned compliance data pipelines, multi-tenant schemas, and async processing systems for 100K+ housing units. The 8yr + 1yr LLM profile Chamberlain requires.
- **Built RabbitMQ-based async system** with retry logic and idempotency guarantees — owned through production incidents; distributed system patterns directly applicable to AI workflow orchestration.
- **Optimized 200+ SQL queries (30% gain)**, owned multi-tenant PostgreSQL schemas for eligibility, financial, and document workflows; 100% PHI compliance on AWS S3 migration (zero security incidents).
- Shipped 20+ full-stack modules across 120+ Agile sprints; microservices with Docker — 50% scalability improvement, 40% deployment time reduction.

PROJECTS

Leasel: 8-Agent LangGraph System with MCP + Tool Integration LANGGRAPH MCP

Python · LangGraph · MCP · FastAPI · TF-IDF + Semantic RAG · ChromaDB · OCR | github.com/venkatalakshmiparimala/Lease_complaince_rag

- 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) — each agent has scoped tool calls and durable memory via graph state; MCP wires tool interfaces cleanly across agents; conditional routing = planning layer determining which agents to invoke per ticket type.
- RAG: section-aware chunking → hybrid TF-IDF + semantic retrieval → reranking → LLM reasoning with citation grounding. HITL escalation when confidence below threshold.
- Eval: 93.6% precision, 87.9% F1 across 10 compliance rules on 1,240 documents; pass@3 88.3%.

AgentProbe: Agentic AI Evaluation + Observability Infrastructure EVAL FRAMEWORK CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

- Trace-level observability + 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression harness with golden datasets, model routing across OpenAI/Anthropic/Groq/Ollama. "Built the harness that tells you when your agents are working and when they're not."

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Agentic AI: LangChain, LangGraph, MCP (Model Context Protocol), Multi-Agent Orchestration, Tool Calling, Tool Integration, RAG Pipelines, Hybrid Search, Semantic Search, Reranking, Embeddings, ChromaDB, Qdrant, FAISS-adjacent, Durable State Management, HITL, Conditional Routing, Agent Reasoning, Agent Memory, LLM-as-Judge, Hallucination Detection, Model Routing, LLMOps, Agent Observability, Trace-Level Observability

Languages & Frameworks: Python (8+ yrs), JavaScript, TypeScript, Java, SQL, R, Bash — FastAPI, Django, Spring Boot, REST APIs

Cloud & Infra: AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock, SageMaker), Azure (Databricks, ADF, ADLS, AML), Docker, Kubernetes, Terraform, Pulumi, CI/CD (GitHub Actions), Prometheus, Grafana, OpenTelemetry

Data & Backend: PostgreSQL, Elasticsearch, Redis, RabbitMQ, Apache Kafka, Celery, Temporal, Distributed Systems, Multi-Tenant Architecture, Microservices, Idempotency, PyTorch, HuggingFace, scikit-learn